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Charlotte Alejandra Guajardo Martínez

Santiago, Chile • charlotte.guajardo@ug.uchile.cl • (+569) 3126 7676

EDUCATION

Universidad de Chile — Santiago, Chile
MSc. Physics — Present

- Relevant coursework: **Graduate level cosmology**, Computational Inteligence.

Universidad Técnica Federico Santa María — Santiago, Chile
BSc. Astrophysics — Mar 2025

- Thesis: *Exploring environmental influences on galaxies through the Smooth Filaments Simulations.*
- Advisor: Rory Smith.

RESEARCH EXPERIENCE

Universidad de Chile, FCFM, Physics Department. — Santiago, Chile
Thesis project, Prof. Domenico Sapone. — Aug. 2025 – Present

- Constructed clustering observables from N-body simulations with their respective covariance matrix.
- Built and trained neural networks to distinguish between cosmological models using power spectra and growth rate measurements.
- Applied interpretability tools for machine learning in the neural network classification, obtaining model-agnostic weights for the neural network decision-making process, which reflect theoretical intuition of the models.

Universidad Federico Santa María, Physics Department. — Santiago, Chile
Thesis project, Prof. Rory Smith. — Nov. 2023 – Oct. 2025

- Generated high-resolution cosmological simulations to study galaxy evolution in the filamentary environment of the cosmic web.
- Analysed the cosmic structure using machine learning tools, obtaining a quantifiable relation between the material loss of galaxies and their relative position to filamentary structure.

ACADEMIC SERVICE

Community

Cosmology in Chile web page, part of the administration team. — March 2026 - Present

Courses organization

Machine Learning Interpretability Techniques — January 2026

CONFERENCES

- Model selection applied to neural networks with clustering data from simulations. Poster presentation delivered at COSMOSUR VIII, Valparaiso, Chile, October 2026.
- *Interpretable Machine Learning applied to growth rate measurements.* Poster presentation delivered at the Physics Department's POST-ERDAY, Santiago, Chile, January 2026.
- *Exploring environmental influences on galaxies through the Smooth Filaments Simulations.* Poster presentation delivered at SOCHIAS XX 2025, Puerto Montt, Chile, October 2025.
- *Exploring environmental influences on galaxies through the Smooth Filaments Simulations.* **Oral presentation** delivered at CECFA 2024, Valparaíso, Chile, October 2024.

TEACHING EXPERIENCE

Universidad de Chile, Physics Department — Santiago, Chile

Teaching assistant, Thermodynamics.	Aug. 2026 – Present
Teaching assistant, Introduction to Modern Physics.	Mar. 2026 – Jul. 2026
Grader, Introduction to Classical Physics.	Mar. 2026 – Jul. 2026
Teaching assistant, Introduction to Modern Physics.	Ago. 2025 – Dic. 2025

Universidad Federico Santa María, Physics Department — Santiago, Chile

Teaching assistant, **General Relativity**.	Mar. 2025 – Jul. 2025
Teaching assistant, Modern Physics.	Mar. 2025 – Jul. 2025
Grader, Modern Physics.	Ago. 2024 – Dic. 2024

SKILLS

Programming languages: Python, Bash/Shell, Markdown, SQL.
Cosmology: CAMB, MG-CAMB.
Machine Learning: Pytorch, Tensorflow, LIME, SHAP.
Numerical simulations: RAMSES, MUSIC, DICE, 1DREAM, GF.
Astronomy: Astropy, TOPCAT, SExtractor, GALFIT, Magphys, SPLAT-VO.
Systems: Linux (Multiple distributions), Git, SSH.
Languages: English (fluent), Spanish (Native).